

WIRTINGER'S INEQUALITY

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INTRODUCTION

I only recently learned that Wirtinger's Inequality, which provides an upper bound on the square integral of a C^1 function f (of average value zero) over $[0, T]$, can be used to prove the Isoperimetric Inequality. This was shown by Hurwitz in 1901: [Hur01]. This article proves the Wirtinger inequality using Parseval's Theorem, but readers should note that there are multiple ways to prove the result. I do not go into the proof of the Isoperimetric Inequality to keep the article brief; one may invoke Green's Theorem to obtain a clean proof in $\mathbb{R}^2$: [Wik26].

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PROOF

Wirtinger's Inequality.

Let f be C^1 over the interval $[0, T]$ with $\int_0^T f(t) dt = 0$, and $f(0) = f(T)$, then we have the estimate

$$\int_0^T |f(t)|^2 dt \leq \frac{T^2}{4\pi^2} \int_0^T |f'(t)|^2 dt.$$

Proof: We note that $\hat{f}(0) = \frac{1}{T} \int_0^T f(t) dt = 0$. Moreover, the Fundamental Theorem of Calculus implies

$$\hat{f}'(0) = \frac{1}{T} \int_0^T f'(t) dt = \frac{1}{T} (f(T) - f(0)) = 0.$$

Suppose $n \neq 0$. Then the Fourier coefficients of f are related to those of f' by the following calculation:

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$$\begin{aligned}
\hat{f}(n) &= \frac{1}{T} \int_0^T f(t) e^{-\frac{2\pi i n t}{T}} dt \\
&= \frac{1}{T} \left(\frac{T}{2\pi i n} \left[f(t) e^{-\frac{2\pi i n t}{T}} \right]_0^T + \frac{T}{2\pi i n} \int_0^T f'(t) e^{-\frac{2\pi i n t}{T}} dt \right) \\
&= \frac{1}{2\pi i n} \int_0^T f'(t) e^{-\frac{2\pi i n t}{T}} dt \\
&= \frac{T}{2\pi i n} \left(\frac{1}{T} \int_0^T f'(t) e^{-\frac{2\pi i n t}{T}} dt \right) \\
&= \frac{T}{2\pi i n} \hat{f}'(n).
\end{aligned}$$

Therefore, for $n \neq 0$ we have

$$|\hat{f}(n)|^2 = \frac{T^2}{4\pi^2 n^2} |\hat{f}'(n)|^2 \leq \frac{T^2}{4\pi^2} |f'(n)|^2.$$

Invoking Parseval's Identity (recall that $\hat{f}(0) = \hat{f}'(0) = 0$), we obtain

$$\begin{aligned}
\frac{1}{T} \int_0^T |f(t)|^2 dt &= \sum_{n=-\infty}^{\infty} |\hat{f}(n)|^2 = \sum_{n \neq 0} |\hat{f}(n)|^2 \leq \frac{T^2}{4\pi^2} \sum_{n \neq 0} |\hat{f}'(n)|^2 \\
&\leq \frac{T^2}{4\pi^2} \left(\frac{1}{T} \int_0^T |f'(t)|^2 dt \right),
\end{aligned}$$

so we obtain the final result that

$$\int_0^T |f(t)|^2 dt \leq \frac{T^2}{4\pi^2} \int_0^T |f'(t)|^2 dt,$$

completing the proof. ■

Notice: Equality occurs in Wirtinger's Inequality if and only if by Parseval's Identity we have that

$$\sum_{n \neq 0} |\hat{f}(n)|^2 - \frac{T^2}{4\pi^2} |\hat{f}'(n)|^2 = 0.$$

Since $|\hat{f}'(n)|^2 = \frac{4\pi^2 n^2}{T^2} |\hat{f}(n)|^2$, the equation above holds if and only if $|\hat{f}(n)| = 0$ for all $n \neq 1$. That is, if and only if f is a linear combination of the functions $e^{-\frac{2\pi i t}{T}}$ and $e^{\frac{2\pi i t}{T}}$.

We also have another result:

Theorem.

If f is as above and g is C^1 and T -periodic,

$$\left| \int_0^T \overline{f(t)} g(t) dt \right|^2 \leq \frac{T^2}{4\pi^2} \int_0^T |f(t)|^2 dt \int_0^T |g'(t)|^2 dt.$$

Proof: By Parseval's Identity, as $\hat{f}(0) = 0$,

$$\left| \int_0^T \overline{f(t)} g(t) dt \right|^2 = T^2 \left| \sum_{n=0} \overline{\hat{f}(n)} \hat{g}(n) \right|^2.$$

Applying the Cauchy-Schwarz Inequality yields the following calculation:

$$\begin{aligned} \left| \int_0^T \overline{f(t)} g(t) dt \right|^2 &\leq T^2 \left(\sum_{n \neq 0} |\hat{f}(n)|^2 \right) \left(\sum_{n \neq 0} |\hat{g}(n)|^2 \right) \\ &= T^2 \left(\sum_{n=-\infty}^{\infty} |\hat{f}(n)|^2 \right) \left(\sum_{n \neq 0} \frac{T^2}{4\pi^2 n^2} |\hat{g}'(n)|^2 \right) \\ &\leq T^2 \left(\sum_{n=-\infty}^{\infty} |\hat{f}(n)|^2 \right) \left(\frac{T^2}{4\pi^2} \sum_{n=-\infty}^{\infty} |\hat{g}'(n)|^2 \right) \\ &= T^2 \left(\frac{1}{T} \int_0^T |f(t)|^2 dt \right) \left(\frac{T}{4\pi^2} \int_0^T |g'(t)|^2 dt \right), \end{aligned}$$

which is to say

$$\left| \int_0^T \overline{f(t)} g(t) dt \right|^2 \leq \frac{T^2}{4\pi^2} \int_0^T |f(t)|^2 dt \int_0^T |g'(t)|^2 dt,$$

the desired inequality. ■

REFERENCES

- [Hur01] Adolf Hurwitz. “Sur le problème des isopérimètres”. In: *Comptes Rendus des Séances de l'Académie des Sciences* 132 (1901). JFM 32.0386.01, pp. 401–403. URL: <https://gallica.bnf.fr/ark:/12148/bpt6k30888.f429>.
- [Wik26] Wikipedia contributors. *Wirtinger's inequality for functions*. 2026. URL: https://en.wikipedia.org/wiki/Wirtinger's_inequality_for_functions (visited on 08/29/2026).